

DNA/RNA Radiolabeling Protocol

Set up labeling reaction:

X ul DNA or RNA (10 pmoles)
2.5 ul 10x PNK buffer
0.5 ul PNK enzyme
1.5 ul P32-gamma-ATP
dH2O (DEPC for labeling RNA) to 25 ul

CasX TS/NTS with non-hydrolysable spacers:

TS:

5'-CGCTAGCTACG **T*T*T*G*A*T*T*T*C*T*G*C*T*G*C*A*G*G*A*TGA**ATCCCGTAATCGCGC-3'

MW: 15664.2g/mol
Concentration: X μ M

*= phosphothioate, red = PAM, green = spacer

For 10pmol of TS: X μ L of stock

NTS:

5'- GCGCGATTACGGGAT **TCAT*C*C*T*G*C*A*G*C*A*G*A*A*A*T*C*A*A*A***CGTAGCTAGCG-3'

MW: 15749.3 g/mol
Concentration: X μ M

*= phosphothioate, red = PAM, green = spacer

For 10pmol of NTS: X μ L of substrate

Add the ATP to empty tubes in the hot room, mix the DNA/buffer/enzyme/H₂O at bench, and then add the DNA/enzyme mixture to the ATP-filled tubes in the hot room.

Incubate @ 37°C for 30 min.

Heat inactivate the PNK @ 65°C for 20 min.

Prepare G25 columns (from GE, green box): vortex thoroughly, twist cap ¼ turn, snap off bottom, spin for 1 min @ 3000 rpm to get rid of liquid. Add 50 ul H₂O to a labeled epi tube, place G25 column in it.

Add 25 ul H₂O to each labeling reaction after heat inactivation is done. Apply entire reaction (now 50 ul total) to G25 column resin. Spin for 2 min @ 3000 rpm.

Since 50 ul H₂O were in bottom of tube and you add your 50 ul reaction, you end with up to 100 ul of 100 nM labeled DNA/RNA.

Measure 1 ul of each reaction with the black rad counter on shelf to get cpm readings.

*TS

XX ul DNA or RNA (10 pmoles)
2.5 ul 10x PNK buffer
0.5 ul PNK enzyme
1.5 ul P32-gamma-ATP
XX µL dH₂O (DEPC for labeling RNA) to 25 ul

*NTS

XX ul DNA or RNA (10 pmoles)
2.5 ul 10x PNK buffer
0.5 ul PNK enzyme
1.5 ul P32-gamma-ATP
XX µL dH₂O (DEPC for labeling RNA) to 25 ul